MIL-H-26666D
25 March 1966
SUPERSEDING
MIL-H-26666C(USAF)
6 October 1965

MILITARY SPECIFICATION

HOSE ASSEMBLY, PNEUMATIC, HIGH PRESSURE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a pneumatic hose assembly having a nominal working pressure of 4,000 psig in the operating temperature range from -65° to +160°F.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-H-775	Hose, Rubber or Fabric (Including Tubing); and Fittings, Nozzles and Strainers, packaging of
MIL-R-3065	Rubber, Fabricated Parts

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-831	Test Reports, Preparation Of
MS28760	Fitting End, Attachable, Hydraulic and Pneumatic, High Pressure Hose (3,000 PSI) Flared Tube
MS33656	Fitting End, Standard Dimensions for Flared Tube Connection and Gasket Seal

Air Force-Navy Aeronautical

AN818	Nut, Coupling
AND10064	Fittings - Installation of Flared Tube, Straight Threaded Connectors

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on date of invitation for bids or request for proposal shall apply.

FSC 4720

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:06Z
Check the source to verify that this is the current version before use.

American Society for Testing and Materials

ASTM Standard D-1248 Polyethylene Molding and Extrusion Materials, Specification for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials

3.3.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them.

3.3.2 Rubber goods. Rubber goods utilized, such as seals, covers, and friction layers, shall be fabricated from rubber components conforming to MIL-R-3065.

3.3.3 Other materials. Materials not otherwise specified herein shall conform to applicable specifications and to the requirements specified herein. All materials that are not specifically described shall be of the highest quality and suitable for the purpose intended.

3.4 Design and construction. The hose assembly shall be designed for a pneumatic working pressure of 4,000 psig and a minimum burst pressure of 16,000 psig.

3.4.1 Reliability. The hose assembly shall have a minimum mean-time-between failures (MTFB) of 870 cycles (0 to 4,000 to 0 psi) with a minimum reliability of 97 percent for a 20-cycle (0 to 4,000 to 0 psi) mission at a confidence factor of 0.90 (see 6.3.1 and 6.3.2).

3.4.2 Hose. The hose shall be constructed to meet the requirements specified herein and to retain the end fittings without slipping or leaking when tested as specified herein. The hose shall consist of an inner tube, reinforcement, and outer cover.

3.4.2.1 Inner tube. The inner tube shall consist of a high-density, polyethylene compound of seamless construction and uniform gage. It shall be compounded and processed to permit the least amount of effusion of compressed gases consistent with the other requirements specified herein. The uncompounded polyethylene material shall conform to ASTM Standard D-1248, type 3. The inner tube shall have a smooth bore, shall be free from pitting and other defects, and shall be cleaned free of dirt, foreign material, or mandrel lubricants.

3.4.2.2 Reinforcement. The reinforcement shall consist of two or more spirally wrapped or braided (or combination thereof) plies of stainless-steel wires. Reinforcement separation materials, if used, shall permit gas effusion from the inner tube to escape to the atmosphere. There shall be no broken or spliced reinforcing wires nor shall any wires be omitted in any length of hose.

3.4.2.3 Outer cover. A cover at least 1/16 inch thick shall be furnished on the

Source: https://assist.dla.mil -- Downloaded: 202610-20T10:06Z
Check the source to verify that this is the current version before use.

outside of the hose. The cover shall consist of an abrasion-, oil-, and ozone-resistant chloroprene polymer, or equivalent synthetic material compounded to meet the requirements specified herein. The cover need not be bonded to the outermost reinforcement ply, but shall not be free to slip or turn along the length of hose. The cover shall permit gas that may effuse from the inner tube to escape to the atmosphere.

3.4.2.4 Diameter. The inside and outside diameters of the hose shall be within the limits shown on figure 1 as applicable to the hose size specified by the procuring activity (see 6.2).

3.4.2.5 Length. The hose assembly shall be furnished in lengths as specified by the procuring activity (see 6.2). The hose assembly length shall be measured as shown on figure 1.

3.4.3 End fittings. The hose assembly end fittings shall conform to the general configuration shown on MS28760, shall be made of stainless-steel, and shall mate with the fitting conforming to MS33656 corresponding to the hose size. The swivel nut dimensions across flats shall conform to an applicable size of AN818. Provisions for wrenching by a hex or flats on fitting body shall be optional. The nipple bore diameter shall conform to the limits shown on figure 1. The end fitting shall be designed for a working pressure of 4,000 psig and a minimum burst pressure of 16,000 psig. The end fittings shall be retained on the assembly without slipping or leaking when coupled to the hose and tested as specified herein.

3.4.3.1 Jump sizes. When specified (see 6.2), the hose assembly shall be supplied with an end fitting of a different nominal size than the hose itself. In such cases, the end fitting swivel nut and nipple shall mate with fittings conforming to MS33656 for the specified jump size while the fitting body and nipple bore diameter shall conform to the applicable dimensions for the specified nominal hose size.

3.4.3.2 Coupling guard. When specified (see 6.2), a coupling guard shall be securely attached to each end of the hose assembly. The coupling guard shall protect the end portions of the hose assembly as necessary to minimize hose failure due to bending at or near the coupling.

3.5 Performance

3.5.1 Proof pressure. The hose assembly shall withstand a hydrostatic pressure of 8,000 psig.

3.5.2 Leakage. The hose assembly shall not leak when subjected to a pneumatic pressure of 4,000 psig.

3.5.3 Cold temperature flexing. The hose assembly shall withstand flexing through 180° at -65° ±2°F.

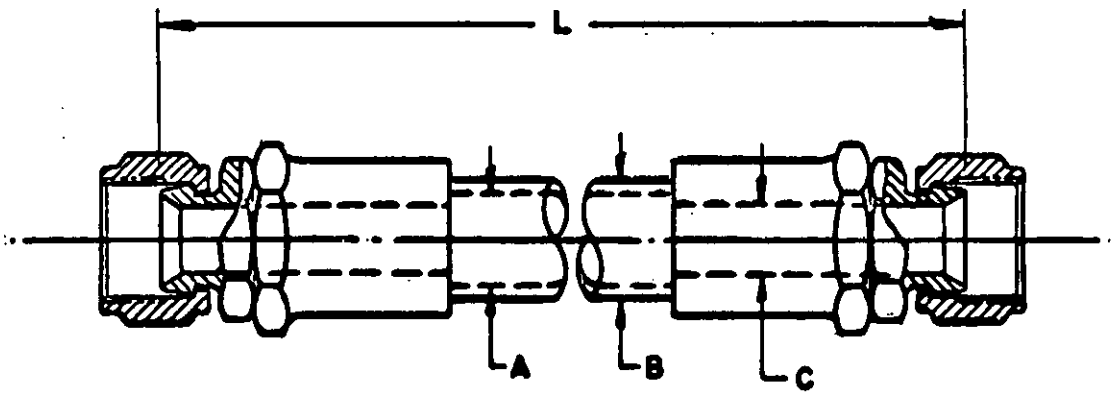
3.5.4 Elongation and contraction. The hose assembly shall withstand pressurization to 4,000 psig without a change in its free length greater than ±3 percent.

3.5.5 Tensile strength. The hose assembly shall withstand a tensile pull of 1,000 pounds while pressurized to 4,000 psi.

3.5.6 Burst pressure. The hose assembly shall withstand, without bursting, pressurization to 16,000 psig.

3.5.7 Overtightening torque. The fittings specified in 3.4.3 shall withstand, without failure or deformation, an overtightening

Source: https://assist.dla.mil -- Downloaded: 202610-20T10:06Z
Check the source to verify that this is the current version before use.



TO MATE WITH MS33656 DASH SIZE	NOMINAL O.D. OF RIGID TUBE (REF)	HOSE INSIDE DIA MINIMUM "A"	HOSE OUTSIDE DIA MAXIMUM "B"	COUPLING INSIDE DIA MIN. "C"	HOSE MIN. INSIDE BEND RAD
-4	1/4	0.234	0.703	0.125	3-1/4
-6	3/8	0.297	0.859	0.188	5-1/4
-8	1/2	0.422	1.000	0.297	6
-10	5/8	0.547	1.218	0.391	6-3/4
-12	3/4	0.672	1.375	0.547	8

NOTES:

1. ALL DIMENSIONS IN INCHES
2. LENGTH OF "L" TO BE WITH +3% OR -1% OF SPECIFIED NOMINAL VALUE

FIGURE 1. HOSE DETAILS

Source: https://assist.dla.mil -- Downloaded: 202610-20T10:06Z
Check the source to verify that this is the current version before use.

torque of 1-3/4 times the maximum values specified on AND10064 for hose end fittings and hose assemblies.

3.5.8 Pressure impulse cycling. The hose assembly shall withstand 2,000 pressure impulse cycles when tested in accordance with 4.6.10.

3.6 Identification of product. The hose assembly shall be identified by a permanently snug-fitting aluminum or stainless-steel band around the hose near the end fitting. Where the hose assembly exceeds 4 feet in length, a band shall be attached near each end fitting of the assembly. The metal band shall be marked in raised, etched, or stamped lettering with the following information appropriately identified:

a. Number and title of this specification

b. Date of assembly in month and year

c. The rated working pressure in psi, followed by the word PNEUMATIC

d. Manufacturer's name and trademark

e. Manufacturer's part number

f. Hose manufacturer's federal supply code number if different from hose assembly manufacturer

g. Federal class code number (FSC 4720) followed by the federal item identification number if available. If not available, follow the class code number with NSL.

3.7 Workmanship. Workmanship shall be of the quality necessary to produce hose assemblies free from all defects which will affect proper functioning in service.

3.8 General cleaning instructions. Following completion of fabrication and assembly operations, the hose assembly shall be thoroughly cleaned. Metal chips, hose cuttings, and other foreign material shall be removed during and after final assembly.

3.8.1 Unless otherwise specified (see 6.2), the hose assembly shall be cleaned in a manner to assure compliance with the cleaning effectiveness requirements of 4.6.11. The cleaning methods and procedures shall be optional. The hose assembly shall not be adversely affected by the cleaning method or procedures.

3.8.1.1 Fluids for cleaning and flushing. Fluids used for cleaning and flushing or as a carrier for cleaning solutions shall be clean and grease-free and shall contain not more than 50 ppm of suspended or dissolved impurities. Water used for flushing or rinsing shall be demineralized and shall have a specific resistance of at least 50,000 ohms. Fluids or solutions that would react chemically with any part of the hose assembly shall not be used.

3.8.1.2 Gas used for purging and drying purposes shall be clean, dry, oil-free air or nitrogen having a dewpoint at 0 psig of not more than 30°F and shall contain not more than 3 parts per million by volume of total hydrocarbons expressed as methane equivalent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:06Z
Check the source to verify that this is the current version before use.

to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the hose assembly shall be classified as follows:

a. Preproduction testing . . . . . .See 4.4

b. Acceptance tests . . . . . .See 4.5

4.3 Test conditions. The dimension inspection specified in 4.6.8 shall be conducted on samples, complete with the outer cover, after cutting hose to length but before attaching end fittings. All other tests shall be conducted on complete hose assembly samples. Unless otherwise specified, tests shall be conducted under ambient conditions.

4.3.1 Instrumentation

4.3.1.1 Pressures. Pressures and pressure differentials shall be measured by means of Bourdon-type gages or other devices capable of complying with 4.3.2.3.1 Pressures shall be recorded in pounds per square inch gage.

4.3.1.2 Temperatures. Temperatures shall be measured by appropriately located thermometers or thermocouples used with calibrated potentiometers. Temperatures shall be recorded in degrees Fahrenheit.

4.3.2 Test data

4.3.2.1 Correction of data. If the tests specified herein cannot be conducted under specified conditions, the tests may, upon approval by the procuring activity, be conducted under other conditions and the performance under the specified conditions calculated from the test results obtained. The manner of calculation shall be demonstrated and actual test data proving the correctness of the calculation methods used shall be presented to the procuring activity.

4.3.2.2 Operational data. Operational data shall be recorded as specified under the individual tests.

4.3.2.3 Tolerances

4.3.2.3.1 Data on gage pressure shall be accurate to within 1 percent of the value reported.

4.3.2.3.2 Data on temperature shall be accurate to within 2°F.

4.3.3 Inspection, servicing, and adjustments. Inspection, servicing, and adjustments shall be performed as specified by the manufacturer unless such instructions are contrary to those necessary for compliance with the requirements specified herein.

4.3.4 Observations. During the progress of all tests, the performance of the hose assembly shall be observed to determine the degree of compliance with all requirements specified herein.

4.3.5 Preparation for test. The hose assembly shall be prepared for tests as specified herein.

4.3.6 Pressure test record. A record of the pressures at which each hose assembly is tested and the length of time during which it is subjected to the pressure shall be retained for examination by the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:06Z
Check the source to verify that this is the current version before use.

4.4 Preproduction testing (see 6.2)

4.4.1 Test samples. Six hose assemblies, 24 ±6 inches long, shall be fabricated from random samples of the bulk hose and end fittings using production processes and procedures and subjected to the preproduction tests. Samples for tensile and burst tests may be reduced to 12 inches long if necessary because of test equipment limitations. Allocation of test samples and testing shall be in accordance with the following schedule:

	Test	Sample Number 1	2	3	4	5	6
4.6.8	Dimension inspection	X	X	X	X	X	X
4.6.1	Examination of product	X	X	X	X	X	X
4.6.2	Proof pressure	X	X	X	X	X	X
4.6.3	Leakage	X	X	X	X	X	X
4.6.4	Cold temperature flexing	X	X				
4.6.5	Elongation and contraction	X	X				
4.6.6	Tensile strength			X	X		
4.6.7	Burst pressure			X	X		
4.6.9	Overtightening torque	X	X				
4.6.10	Pressure impulse cycling					X	X.

4.4.2 Test report. After completion of the preproduction tests, a test report shall be prepared in accordance with MIL-STD-831.

4.4.2.1 Reliability and maintainability information. The following information shall be included as an appendix to the test report:

a. All failures, servicing, adjustments, maintenance, and irregular functioning shall be identified by accumulated operating time, cycles, miles, or position in the test procedure, as appropriate. Test conditions at the time of the events identified shall be recorded

b. A summary of the engineering analysis and of any tests conducted to determine assignable causes for any failure or irregular functioning

c. A summary of the engineering analysis leading to any corrections made to design, construction, quality control, or other procedures, or leading to any corrections to be made to production items or proposed to be made. The summary shall also include an analysis of the predicted effectiveness of such corrections

d. Clock time and man-hours required for each maintenance and servicing action taken during the tests. A brief description of the experience and qualifications of the personnel taking such actions

e. Test activity or contractor comments on item features or requirements that, if modified, should improve the item

f. Test activity or contractor comments on use or maintenance conditions to be avoided or cultivated to increase the reliability or useful life of the item

g. Any of the above information that is already included in the test report body need not be repeated in the information required by this paragraph, but clear reference to the location of the data shall be included.

4.5 Acceptance tests. The acceptance tests shall consist of the following:

a. Individual tests . . . . . .See 4.5.1

b. Sampling tests . . . . . .See 4.5.2.

4.5.1 Individual tests. Each hose assembly shall be subjected to the following tests

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:06Z
Check the source to verify that this is the current version before use.

as described under 4.6, conducted in the order listed:

a. Examination of product . . .See 4.6.1

b. Proof pressure . . . . . See 4.6.2

c. Leakage . . . . . See 4.6.3.

4.5.2 Sampling tests

4.5.2.1 Lot. A lot shall consist of hose assemblies manufactured under essentially the same conditions and submitted for inspection at substantially the same time.

4.5.2.2 Sampling plan. For each lot of five hundred hose assemblies or fraction thereof produced, four hose assembly samples, 18 ±6 inches long, shall be fabricated from random samples of the bulk hose and end fittings using production processes and procedures. Allocation of test samples and testing shall be in accordance with the following schedule. In case any sample fails to fulfill the specified requirements, two additional samples from the same lot shall be tested in the same manner.

	Test	Sample Number 1	2	3	4
4.6.8	Dimension inspection	X	X	X	X
4.6.2	Proof pressure	X	X	X	X
4.6.3	Leakage	X	X	X	X
4.6.6	Tensile strength	X	X		
4.6.7	Burst pressure	X	X		
4.6.10	Pressure impulse cycling			X	X.

4.5.2.2.1 Rejection and retest. If more than one of the four original samples or either of the two additional samples fail to fulfill the specified test requirements, the entire lot shall be rejected.

4.5.2.2.2 Individual tests may continue. For production and operational reasons, individual tests or other sampling plans may be continued pending the investigation of a sampling test failure. Final acceptance of the entire lot or hose assemblies on hand or produced later, whichever the case may be, shall not be made until it is determined that all hose assemblies meet all the requirements of the specification.

4.5.3 Defects in hose assemblies already accepted. The investigation of a test failure could indicate that defects may exist in hose assemblies already accepted. If so, the producer shall fully advise the procuring activity of all the defects likely to be found and the method of correcting them.

4.6 Test methods

4.6.1 Examination of product. The hose assembly shall be visually examined to determine compliance with the requirements specified herein with respect to materials, workmanship, length, and marking.

4.6.2 Proof pressure. The hose assembly shall be subjected to 8,000 psig hydrostatic pressure for not less than 30 seconds nor more than 5 minutes. Clean water shall be used as the pressurizing fluid. Any indication of leakage, distortion, or other malfunction shall be cause for rejection. Preproduction and sampling plan items shall be subjected to not less than 160°F for 1 hour immediately prior to conducting this test.

4.6.3 Leakage. The hose assembly shall be pressurized to 4,000 psig pneumatic pressure with the assembly under water. The pressure shall be maintained for not less than 1 minute nor more than 15 minutes. Indication of rapid forming and breaking away of bubbles from any point on the outer surface of the hose or any part of

Source: https://assist.dla.mil -- Downloaded: 202610-20T10:06Z
Check the source to verify that this is the currentversion before use.

the end fitting shall be cause for rejection. In the event of a dispute, the effusion from any 12-inch section of the hose assembly shall be collected in an inverted graduate for 15 minutes. At the end of 15 minutes, the amount of gas accumulated in the graduate shall not exceed 50 cc.

4.6.4 Cold temperature flexing. The hose assembly shall be placed in a cold chamber and soaked at -65° ±2°F for 24 hours. At the end of the soaking period, and with the assembly still at the specified temperature, the assembly shall be flexed through 180° to the applicable bend radius shown on figure 1 and then straightened. The assembly shall again be flexed through 180° to the same bend radius in such a direction that the portion of the hose under maximum compression during the first flexing will be under maximum expansion during the second flexing. The sample shall be flexed in both directions within 4 seconds. The hose assembly shall be visually examined, and then subjected to the leakage test specified in 4.6.3. Any hose assembly showing evidence of cracking, splitting, distortion, or leakage shall be considered to have failed the test.

4.6.5 Elongation and contraction. The hose assembly shall be placed in a straight position with one end unrestrained. Before applying pressure, the free length of the assembly shall be measured. The assembly shall then be pressurized to 4,000 psig and maintained at that pressure for 5 minutes. At the end of 5 minutes and while still at the test pressure, the change in free length shall be measured and shall not exceed +3 percent (elongation) or -3 percent (contraction) for all sizes.

4.6.6 Tensile strength. The hose assembly shall be placed in a straight position and pressurized to 4,000 psig using a suitable liquid. While pressurized, the hose shall be pulled by the end fittings to 1,000 pounds tensile pull at the minimum rate of 1 inch per minute. Any failure below 1,000 pounds pull shall be cause for rejection.

4.6.7 Burst pressure. The hose assembly shall be placed in a straight position with one end unrestrained. The assembly shall be pressurized with a suitable fluid. The rate of pressure rise shall be 20,000 ±5,000 psi per minute until failure occurs. If any component of the hose assembly bursts or cracks at less than 16,000 psi, it shall be cause for rejection. The hose assembly shall be subjected to 160°F for 1 hour immediately prior to conducting this test.

4.6.8 Dimension inspection. The hose shall be checked to determine compliance with the dimension requirements specified in 3.4.2.4. A hose found to exceed the tolerance limits specified shall be cause for rejection.

4.6.9 Overtightening torque. The flared end fittings of the hose assembly shall be assembled on a fitting of steel construction conforming to MS33656. The fitting shall be tightened to an overtight-torque of 1-3/4 times the maximum values specified on AND10064 for hose end fittings and hose assemblies, and loosened. The tightening and loosening sequence shall be repeated until a total of not less than 15 cycles have been performed. The end fitting shall then be removed and examined for damage. Evidence of failure, thread damage, or deformation of the end fitting assembly, or difficulty in turning the swivel nut on the nipple by hand shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 202640-20T10:06Z
Check the source to verify that this is the currentversion before use.

4.6.10 Pressure impulse cycling. The hose assembly shall be subjected to the following tests:

4.6.10.1 Ambient temperature pressure impulse cycling. The hose assembly shall be fixed in a semicircular position with the longitudinal axes of the end fittings parallel. The hose assembly shall be subjected to a hydraulic or pneumatic pressure impulse cycle from 0 to 4,000 to 0 psig at the rate of not less than 60 ±10 cpm for not less than 1,000 cycles. The hose assembly shall be visually examined, and then subjected to the leakage test specified in 4.6.3. Any hose assembly showing evidence of cracking, splitting, distortion, or leakage shall be considered to have failed the test.

4.6.10.2 Elevated temperature pressure impulse cycling. The hose assembly shall be subjected to the pressure impulse cycling and testing specified in 4.6.10.1, except the temperature shall be not less than 160°F. Any hose assembly showing evidence of cracking, splitting, distortion, or leakage shall be considered to have failed the test.

4.6.11 Cleaning effectiveness. One hose assembly shall be selected at random from each lot of two hundred hose assemblies, or fraction thereof, that have been cleaned in accordance with 3.8.1 and subjected to the following test:

4.6.11.1 The hose assembly shall be filled with flushing water conforming to 3.8.1.1 and, while filled, the entire length of hose assembly including end fittings shall be brushed internally with a clean brush having a diameter at least 1/16 inch larger than the inside diameter of the hose.

4.6.11.2 After the brushing operation, the water shall be drained from the hose assembly into a clean beaker or similar container. The water in the beaker shall be vigorously agitated, and a 50 ±5 cc sample transferred to an evaporating dish before any residue can again settle to the bottom of the beaker.

4.6.11.3 The measured sample shall then be evaporated to dryness at 221°F and the residue weighed and microscopically examined. Evidence of particle sizes larger than 100 microns or a total residue weight of more than 1 milligram per milliliter of original measured sample shall be cause for rejection of the lot. If the lot is rejected, the hose assemblies shall be recleaned in accordance with 3.8.1 and a new sample taken for compliance with this test. If the test shows the cleaning methods to be satisfactory, the hose assembly shall be dried with drying gas conforming to 3.8.1.2 and preserved and packaged with the remainder of the lot.

4.6.12 Reliability demonstration and test. Satisfactory completion of all tests specified herein demonstrates compliance with the quantitative reliablilty requirements of this specification.

4.7 Inspection of preparation for delivery. The inspection of the preservation, packaging, packing, and marking shall be in accordance with the requirements of section 5 herein.

5 PREPARATION FOR DELIVERY

5.1 Preservation and packaging, and packing. The hose assemblies shall be preserved, packaged, and packed in accordance with MIL-H-775.

5.2 Marking of shipment. Interior packages and exterior shipping containers

Source: https://assist.dla.mil -- Downloaded: 202610-20T10:06Z
Check the source to verify that this is the currentversion before use.

shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

Hose Assembly, Pneumatic, High Pressure (4,000 PSI).

5.2.1 In addition, the length shall be specified in inches and the jump size end fittings shall be indicated when applicable.

6. NOTES

6.1 Intended use. The hose assembly covered by this specification is intended for use as a component of high-pressure pneumatic ground support equipment used in aircraft ground servicing and engine starting applications. (DO NOT USE FOR OXYGEN.)

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Required dash size of hose assembly (see 3.4.2.4 and figure 1)

c. Required length of hose assembly (see 3.4.2.5 and figure 1)

d. End fitting jump size when required (see 3.4.3.1)

e. When coupling guard is required (see 3.4.3.2)

f. When cleaning level specified is not required (see 3.8.1)

g. Condition and location for preproduction testing (see 4.4)

h. Applicable levels of packaging and packing (see section 5).

6.3 Definitions. For the purpose of this specification, the following definitions will apply:

6.3.1 Reliability. Reliability is defined as the probability of performing a specified function under given conditions without failure for a specified period of time. Recognizing that, in general, the rate of failure of equipment is fairly constant throughout the life of the equipment, the probability of nonfailure over an operating time interval decreases exponentially as a function of the length of the interval, during which time there is a constant failure rate, and can be expressed as follows:

Confidence level: 90 percent

MTBF = Total test time

Reliability = e^{-x} where $x = \frac{\text{Mission Time}}{\text{MTBF}}$

The $2.3 = \frac{4.61}{2}$ is based on the constant for for the Poisson/Chi squared distribution, assuming an exponential (2 degrees freedom) distribution even though the failure rate for the test time is zero.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:06Z
Check the source to verify that this is the current version before use.

6.3.2 Mean-time-between-failures. The mean life or mean-time-between-failures is the arithmetical mean (average) of the operating time between failures.

Custodians:
Air Force - 11
Army - MO
Navy - SH

Preparing activity
Air Force - 11

Project No.4720-0092

Reviewers:
Air Force - 11, 69
Army - MI, WC, MU, MO
Navy - YD, SH
Defense Supply Agency - CS

User:
Navy - MC

Source: https://assist.dla.mil -- Downloaded: 202610-20T10:06Z
Check the source to verify that this is the current version before use.